# config_endpoint

Command: `config_endpoint`

## Description

Using config_endpoint user can add/delete/modify the end point configuration

## Request

### Request Fields

| Field | Type | Required | Description |
|---|---|---|---|
| `command` | string | **Yes** | Command issued to configure active endpoint |
| `requestId` | string | **Yes** | A unique identifier for the request, allowing tracking and debugging of the operation. |
| `epConfig` | object | No | |
| `epConfig.operation` | string (enum) | **Yes** | Specifies the type of operation to perform, either adding or deleting a configuration. |
| `epConfig.configuration` | object | **Yes** | |
| `epConfig.configuration.endpointName` | string | **Yes** | Identifies the name of the endpoint, which determines its purpose or function. |
| `epConfig.configuration.epType` | string (enum) | **Yes** | Specifies the type of the endpoint. |
| `epConfig.configuration.protocol` | string (enum) | **Yes** | Defines the communication protocol used by the endpoint. |
| `epConfig.configuration.activate` | boolean | **Yes** | Indicates whether the endpoint is active or inactive. |

| Field | Type | Required | Description |
|---|---|---|---|
| `epConfig.configuration.url` | string | **Yes** | Specifies the URL associated with the endpoint. |
| `epConfig.configuration.verificationType` | string (enum) | **Yes** | Specifies the type of verification used for establishing secure connections. |
| `epConfig.configuration.port` | integer | **Yes** | Specifies the port number used for communication. |
| `epConfig.configuration.qosCommon` | integer | **Yes** | Defines the Quality of Service (QoS) level for the endpoint. |
| `epConfig.configuration.tenantId` | string | **Yes** | Identifies the tenantID associated with the configuration. |
| `epConfig.configuration.mqttParams` | object | No | |
| `epConfig.configuration.mqttParams.keepAlive` | integer | No | Specifies the keep-alive interval in seconds for MQTT connections. |
| `epConfig.configuration.mqttParams.cleanSession` | boolean | No | Indicates whether to start a clean session for the MQTT client. |
| `epConfig.configuration.mqttParams.reconnectDelayMin` | integer | No | Specifies the minimum delay in seconds before reconnecting. |
| `epConfig.configuration.mqttParams.reconnectDelayMax` | integer | No | Specifies the maximum delay in seconds before reconnecting. |
| `epConfig.configuration.mqttParams.clientId` | string | No | Specifies the client ID used for MQTT connections. |
| `epConfig.configuration.mqttParams.username` | string | No | The username for authentication in MQTT connections. |

| Field | Type | Required | Description |
|---|---|---|---|
| `epConfig.configuration.mqttParams.password` | string | No | The password for authentication in MQTT connections. |
| `epConfig.configuration.mqttParams.publishTopics` | array | No | |
| `epConfig.configuration.mqttParams.subscribeTopics` | array | No | |
| `epConfig.configuration.securityParams` | object | No | Defines the security parameters for the endpoint. |
| `epConfig.configuration.securityParams.format` | string (enum) | **Yes** | Specifies the format of the security certificates. Currently, only PEM is supported. |
| `epConfig.configuration.securityParams.algorithm` | string (enum) | **Yes** | Specifies the algorithm used for encryption. |
| `epConfig.configuration.securityParams.caCertificateFile` | string | **Yes** | Specifies the file containing the CA certificates. |
| `epConfig.configuration.securityParams.clientCert` | string | **Yes** | Specifies the file containing the client certificate. |
| `epConfig.configuration.securityParams.clientKey` | string | **Yes** | Specifies the file containing the client private key. |
| `epConfig.configuration.eventConfiguration` | object | No | |
| `epConfig.configuration.eventConfiguration.antenna` | boolean | No | set to true to enable antenna events |

| Field | Type | Required | Description |
|---|---|---|---|
| `epConfig.configuration.eventConfiguration.terminalConnection` | boolean | No | set to true to enable terminal connection/disconnection events |
| `epConfig.configuration.eventConfiguration.firmwareUpdate` | boolean | No | set to true to enable fw update events |
| `epConfig.configuration.eventConfiguration.gpi` | boolean | No | set to true to enable gpi events |
| `epConfig.configuration.eventConfiguration.network` | boolean | No | set to true to enable network events |
| `epConfig.configuration.eventConfiguration.exceptions` | boolean | No | set to true to enable exception events |
| `epConfig.configuration.eventConfiguration.ntp` | boolean | No | set to true to enable battery alerts |
| `epConfig.configuration.eventConfiguration.userApp` | boolean | No | set to true to enable user app events |
| `epConfig.configuration.eventConfiguration.heartbeat` | boolean | No | set to true to enable heart beat |
| `epConfig.configuration.eventConfiguration.power` | boolean | No | set to true to enable power source alerts |
| `epConfig.configuration.eventConfiguration.battery` | boolean | No | set to true to enable battery alerts |
| `epConfig.configuration.eventConfiguration.temperature` | boolean | No | set to true to enable temperature alerts |

| Field | Type | Required | Description |
|---|---|---|---|
| `epConfig.configuration.eventConfiguration.fileDownload` | boolean | No | set to true to enable file download alerts |
| `epConfig.configuration.eventConfiguration.cpuUsage` | boolean | No | set to true to enable cpu usage alerts |
| `epConfig.configuration.eventConfiguration.flashUsage` | boolean | No | set to true to enable flash usage alerts |
| `epConfig.configuration.eventConfiguration.ramUsage` | boolean | No | set to true to enable ram usage alerts |
| `epConfig.configuration.eventConfiguration.heartbeatConfiguration` | object | No | |
| `epConfig.configuration.eventConfiguration.heartbeatConfiguration.interval` | integer | No | heart beat interval in seconds |
| `epConfig.configuration.eventConfiguration.heartbeatConfiguration.inventoryStatus` | boolean | No | set to true to enable inventory status as part of heartbeat message |
| `epConfig.configuration.eventConfiguration.heartbeatConfiguration.batteryStatus` | boolean | No | set to true to enable battery status as part of heartbeat message |
| `epConfig.configuration.eventConfiguration.heartbeatConfiguration.userApps` | boolean | No | set to true to enable user app status as part of heartbeat message |
| `epConfig.configuration.eventConfiguration.cpuThreshold` | integer | No | cpu usage value(%) above which alert will be triggered |

| Field | Type | Required | Description |
|---|---|---|---|
| `epConfig.configuration.eventConfiguration.ramThreshold` | integer | No | ram usage value(%) above which alert will be triggered |
| `epConfig.configuration.eventConfiguration.flashThreshold` | integer | No | flash usage value(%) above which alert will be triggered |
| `epConfig.configuration.eventConfiguration.temperatureThreshold` | integer | No | temperature treshold value above which alert will be triggered |

## Request Examples

### Add MDM endpoint

```
{
  "command": "config_endpoint",
  "requestId": "1233",
  "epConfig": {
    "operation": "add",
    "configuration": {
      "endpointName": "mgmt_tst",
      "epType": "MGMT",
      "protocol": "MQTT",
      "activate": true,
      "url": "test.example.com",
      "verificationType": "NONE",
      "port": 1883,
      "qosCommon": 1,
      "tenantId": "ZEBRA",
      "mqttParams": {
        "keepAlive": 300,
        "cleanSession": true,
        "reconnectDelayMin": 50,
        "reconnectDelayMax": 500,
        "clientId": "zebra-rfd40-01",
        "username": "username",
        "password": "pass@1234",
        "publishTopics": [
          {
            "topic": "MDM/clients/resp",
            "qos": 1,
            "retain": false
          },
          {
            "topic": "MDM/clients/event",
```

```
            "qos": 1,
            "retain": false
          },
          {
            "topic": "MDM/clients/rfid",
            "qos": 0,
            "retain": true
          }
        ],
        "subscribeTopics": [
          {
            "topic": "MDM/clients/cmnd",
            "qos": 0,
            "retain": false
          }
        ]
      }
    }
  }
}
```

### Delete endpoint

```
{
  "command": "config_endpoint",
  "requestId": "1233",
  "epConfig": {
    "operation": "delete",
    "configuration": {
      "endpointName": "mgmt_tst",
      "epType": "MGMT"
    }
  }
}
```

### Add data endpoint

```
{
  "command": "config_endpoint",
  "requestId": "1233",
  "epConfig": {
    "operation": "add",
    "configuration": {
      "endpointName": "dataEP",
      "epType": "DATA1",
      "protocol": "MQTT",
      "activate": true,
      "url": "17L11-GG6663-1",
      "verificationType": "NONE",
      "port": 1883,
      "qosCommon": 1,
```

```
      "tenantId": "ZEBRA",
      "mqttParams": {
        "keepAlive": 300,
        "cleanSession": true,
        "reconnectDelayMin": 50,
        "reconnectDelayMax": 500,
        "clientId": "zebra-rfd40-01",
        "username": "username",
        "password": "password"
      }
    }
  }
}
```

### Add control endpoint

```
{
  "command": "config_endpoint",
  "requestId": "1233",
  "epConfig": {
    "operation": "add",
    "configuration": {
      "endpointName": "ctrlEP",
      "epType": "CTRL",
      "protocol": "MQTT",
      "activate": false,
      "url": "17L11-GG6663-1",
      "verificationType": "NONE",
      "port": 1883,
      "qosCommon": 1,
      "tenantId": "zebra",
      "mqttParams": {
        "keepAlive": 300,
        "cleanSession": true,
        "reconnectDelayMin": 50,
        "reconnectDelayMax": 500,
        "clientId": "zebra-rfd40-01",
        "username": "mqttuser",
        "password": "password",
        "publishTopics": [
          {
            "topic": "MDM/clients/resp",
            "qos": 1,
            "retain": false
          },
          {
            "topic": "MDM/clients/event",
            "qos": 1,
            "retain": false
          },
          {
            "topic": "MDM/clients/rfid",
            "qos": 0,
```

```
            "retain": true
          }
        ],
        "subscribeTopics": [
          {
            "topic": "MDM/clients/cmnd",
            "qos": 0,
            "retain": false
          }
        ]
      }
    }
  }
}
```

## Response

### Response Fields

| Field | Type | Required | Description |
|---|---|---|---|
| `command` | string | No | |
| `requestId` | string | No | |
| `apiVersion` | string (enum) | No | |
| `response` | object | No | Status of the command execution — check `code` to confirm success (0) or identify the error. |

| Field | Type | Required | Description |
|---|---|---|---|
| `response.code` | integer | **Yes** | Response code indicating success or failure.<br>• **0** — Success<br>• **10** — Configuration already exist<br>• **23** — Invalid enum value<br>• **25** — Max 3 publish topics exceeded<br>• **26** — Max 1 subscribe topic exceeded<br>• **27** — Invalid tenant ID length |
| `response.description` | string | **Yes** | Human-readable description of the response code (e.g. "Success", "Invalid payload"). |

### Response Examples

#### example1

```
{
  "command": "config_endpoint",
  "requestId": "abc123",
  "apiVersion": "V1.1",
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```

## Status / Error Codes

| Code | Status Constant | Description |
|---|---|---|
| 0 | `IOT_STATUS_SUCCESS` | Success |

| Code | Status Constant | Description |
|---|---|---|
| 10 | `IOT_ERROR_CONFIG_ALREADY_EXIST` | Configuration already exist |
| 23 | `IOT_ERROR_INVALID_ENUM` | Invalid enum value |
| 25 | `IOT_ERROR_PUBLISH_TOPICS_EXCEEDED` | Max 3 publish topics exceeded |
| 26 | `IOT_ERROR_SUBSCRIBE_TOPIC_EXCEEDED` | Max 1 subscribe topic exceeded |
| 27 | `IOT_ERROR_INVALID_TENANTID_LENGTH` | Invalid tenant ID length |